

SAVITRIBAI PHULE PUNE UNIVERSITY



A Project Report On

DECENTRALIZED LIBRARY SYSTEM USING BLOCKCHAIN

Submitted Towards The
Partial Fulfillment Of The Requirements Of
BSc in Blockchain Technology

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Abstract

The traditional digital library and content distribution systems rely heavily on centralized servers and third-party intermediaries, which makes them vulnerable to issues such as data breaches, ownership disputes, manipulation of records, and unfair distribution of revenue. These systems often lack transparency, traceability, and user control, leading to inefficiency and trust deficits between creators, publishers, and readers. To address these challenges, this paper proposes the use of blockchain technology to build a decentralized library management system — SmartShelf — that ensures transparency, security, and immutability in digital content transactions.

SmartShelf leverages the Ethereum blockchain and smart contracts to enable secure publishing, purchasing, and access of e-books and digital content without relying on a central authority. The use of blockchain's immutable public ledger ensures that each transaction and ownership record is tamper-proof and verifiable. By eliminating intermediaries, SmartShelf enhances trust, reduces operational costs, and accelerates content distribution while providing fair compensation to authors through automated royalty mechanisms.

The primary objective of this paper is to explore how blockchain technology can transform digital content management by establishing decentralized ownership, preventing piracy, and automating access through smart contracts. This paper discusses the shortcomings of traditional centralized library systems and demonstrates how blockchain-based solutions can overcome these limitations by offering a transparent, secure, and efficient decentralized platform for digital content management.

Acknowledgment

It gives me great pleasure and satisfaction to present the final project report on '**Decetralized Library Platform**'.

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Chapter 1

SYNOPSIS

1.1 Project Title *SMARTSHELF*

A DECENTRALIZED LIBRARY PLATFORM

1.2 Internal Guide

Assistant Prof. Sahil Shaikh

1.3 PROBLEM STATEMENT

Traditional digital library systems that are centralized, easily tampered with, and prone to piracy fail to protect authors' rights and ensure fair revenue distribution. This lack of transparency, ownership verification, and security leads to unauthorized content sharing, data manipulation, and loss of trust among creators and users. To address these challenges, a blockchain-based decentralized library platform is needed. It must ensure transparent royalty distribution, immutable ownership records, secure content access, and efficient verification of digital assets to promote fairness, accountability, and innovation in digital publishing.

1.4 Abstract

The current digital library management system used for publishing and content distribution involves centralized, manual, and third-party-controlled processes that are prone to data manipulation, piracy, and lack of transparency. This conventional method is inefficient and often results in unauthorized sharing, unfair royalty distribution, and inaccurate record keeping. To address these issues, this paper proposes the use of blockchain technology to develop a decentralized digital library platform called SmartShelf, which ensures data integrity, transparency, and security. The blockchain-based SmartShelf model provides an immutable and decentralized public ledger that enhances the accuracy and traceability of digital content ownership. By eliminating intermediaries, SmartShelf can automate royalty distribution, streamline transactions in real time, and reduce the operational costs and risks associated with digital content management.

The primary objective of this paper is to explore how blockchain technology can revolutionize digital publishing by ensuring secure ownership records, enabling transparent revenue tracking, and preventing piracy through cryptographic algorithms and smart contracts. This paper will first discuss the limitations of traditional digital library systems, including security, efficiency, and transparency, and then examine how blockchain-based SmartShelf addresses these challenges through decentralized and automated content management.

1.5 Goals and Objectives

1. To create a system that can be used by any organization or business seeking to digitize and automate their inventory management process.
2. To establish a permanent, tamper-proof record of all product a stock movements within a blockchain network.
3. To enable inventory details and product data to be securely stored, updated, and assigned to a verified user or department account.
4. The SmartShelf system using blockchain is a distributed platform that will record all inventory-related transactions occurring during stock updates, sales, and transfers.

1.6 RELEVANT MATHEMATICS ASSOCIATED WITH THE PROJECT

System Description :

Input:

- Author details, book metadata (title, description, price, file hash/IPFS link)
- User wallet addresses and purchase transaction details
- Smart contract functions for book registration, ownership verification, and royalty distribution

Output:

- Verified and immutable records of book ownership and transactions stored on the blockchain
- Transparent royalty distribution to authors through automated smart contracts
- Secure access control for users to view or download purchased books

Success Conditions

- Successful verification and recording of book ownership and purchase transactions on the blockchain
- Accurate and automated royalty distribution to authors via smart contracts
- Secure storage, retrieval, and traceability of digital book data
- Efficient transaction processing with low latency and minimal gas consumption

Failure Conditions

- Inability to validate or record book transactions, leading to unauthorized access or loss of ownership data
- System downtime, data loss, or corruption affecting the integrity and availability of library information
- Smart contract vulnerabilities resulting in incorrect royalty distribution or security breaches
- Inefficient data handling or poor algorithm optimization causing high gas costs and slow transaction throughput

Chapter 2

TECHNICAL KEYWORDS

2.1 AREA OF PROJECT

Educational Institutions:

Collaborate with universities, colleges, and schools to implement SmartShelf as a blockchain-based decentralized library system. This can help institutions maintain transparent, tamper-proof records of issued, returned, and reserved books while reducing dependency on manual cataloging and paper-based systems.

Public Libraries:

Public libraries can benefit from SmartShelf by adopting a secure digital system to track book availability, borrowing history, and user activity. Through token-based access and smart contracts, libraries can automate fines, membership renewals, and borrowing limits while ensuring accountability and accessibility for all users.

Online Learning Platforms:

Partner with e-learning platforms to integrate SmartShelf for managing digital resources such as e-books, journals, and study materials. This ensures decentralized access control, prevents unauthorized duplication, and provides transparency in resource usage through blockchain technology.

Publishing and Book Distribution Industry:

Collaborate with publishers and book distributors to use blockchain for tracking ownership, copyrights, and circulation of books. SmartShelf can provide a digital ledger ensuring authenticity of book copies and preventing piracy or unauthorized replication of educational content.

Non-Profit and Community Libraries:

Work with NGOs and local community centers that promote education and literacy. SmartShelf can help these organizations manage book donations, track resource utilization, and provide open access to reading materials without the need for costly centralized infrastructure.

Research and Academic Collaboration:

Partner with research institutions working in blockchain, education, or information management. These collaborations can help in enhancing SmartShelf's functionality, experimenting with NFT-based book ownership, and exploring new models of decentralized knowledge sharing.

2.2 TECHNICAL KEYWORDS

1. Blockchain Technology
2. Smart Contracts.
3. Decentralization.
4. Consensus Mechanism.
5. Immutability
6. Cryptography.
7. Peer-to-Peer (P2P).
8. Digital Identity.
- 9.Tokenization.
10. Data Validation.
11. Scalability.
12. User Interface (UI) and User Experience (UX).
13. NFT Integration
14. Distributed Ledger Technology (DLT)
15. Access Control

Chapter 3

INTRODUCTION

3.1 Project Idea

- **Enhanced Security:** SmartShelf uses blockchain encryption and smart contracts to protect digital content and user data from tampering or piracy.
- **Fair Royalty Distribution:** Automated smart contracts ensure transparent and instant payments to authors for every book access or purchase.
- **Transparency & Authenticity:** Every transaction—book issue, return, or ownership—is recorded immutably, ensuring trust among users and creators.
- **Efficiency & Accessibility:** The decentralized system streamlines operations, reduces intermediaries, and provides seamless access to verified digital resources.

3.2 Motivation Of The Project

- The motivation behind developing SmartShelf is to reduce piracy and unauthorized distribution of digital books through blockchain security.
- The system enables transparent validation of book ownership and author rights as all transactions are stored immutably on the public ledger.
- By eliminating intermediaries like third-party publishers, SmartShelf ensures direct and fair royalty payments to authors using smart contracts.
- Blockchain technology in SmartShelf helps lower operational costs, saves time in digital content verification, and builds trust between authors, readers, and publishers.

Chapter 4

PROBLEM DEFINITION AND SCOPE

4.1 PROBLEM STATEMENT

The problems with the current digital library systems include the fact that book distribution and management are centralized, slow, and vulnerable to piracy. There is an increasing number of copyright violations, unfair royalty distribution, and data breaches in existing systems. The involvement of intermediaries such as publishers or hosting platforms often leads to reduced author profits and lack of transparency. One major drawback of the current digital library system is that verifying ownership or tracking digital book access requires extensive manual work, making the process inefficient. Present systems also fail to ensure fair compensation for authors and do not prevent duplication or unauthorized sharing of books. These systems rely on centralized databases, which are prone to manipulation and data loss. Currently, most libraries and e-book platforms store data through centralized servers, which makes them susceptible to hacking and piracy. Moreover, the lack of a transparent verification method makes it difficult to authenticate book ownership, ensure fair payments, and maintain trust between authors, publishers, and readers.

4.1.1 Goals and objectives

1. To create a system that can be used by any institution, library, or platform that wants to digitize and decentralize their book management system.
2. To create a permanent, unalterable record of book ownership and transactions for authors and readers.
3. To enable digital books and ownership rights to be recorded and linked to the user's blockchain account.
4. The SmartShelf system using blockchain is a distributed platform that will store all transactions related to book issuance, borrowing, purchase, and royalty distribution.

4.1.2 Statement of scope

1. The SmartShelf system using blockchain is a distributed platform that will store all transactions related to book issuance, borrowing, purchases, and royalty distribution.
2. This system can be easily adopted by libraries, educational institutions, or digital publishing platforms by creating a standardized framework for decentralized book management, similar to existing digital library solutions but with enhanced security and transparency.
3. This will be helpful for authors, readers, and administrators to track book ownership, automate royalty payments, and manage digital content efficiently, while significantly accelerating the process of book access and record management.

4.2 MAJOR CONSTRAINTS

- Legal and Regulatory Framework: SmartShelf must comply with copyright laws, digital content regulations, and data privacy policies in the jurisdictions where it operates. Any conflicts could result in delays or legal challenges.
- Data Quality: The accuracy and completeness of book metadata, author information, and transaction records are critical. Errors or inconsistencies can lead to incorrect ownership records, disputes, or misallocated royalties.
- Infrastructure: SmartShelf requires robust infrastructure, including servers (for frontend/back-end integration), reliable internet, and blockchain nodes to ensure smooth and secure functioning.
- Human Resources: The project demands skilled personnel with expertise in blockchain development, smart contracts, library management, and UI/UX design to implement and maintain the system effectively.
- Funding: Development, deployment, and maintenance of SmartShelf—including blockchain integration and front-end design—can be costly, making sufficient funding a major constraint.
- Stakeholder Engagement: Success depends on the involvement of authors, publishers, libraries, educational institutions, and users. Lack of engagement may result in low adoption, resistance, or operational inefficiencies.
- Technical Support and Maintenance: Continuous updates, bug fixes, and smart contract audits are necessary. Inadequate technical support can compromise security and system reliability.

4.3 HARDWARE RESOURCES REQUIRED

Hardware Resource	Parameter	Minimum Requirement	Justification
Central Processing Unit (CPU)	Speed (GHz)	2.5 or higher	Solidity compilation, 3D modeling, React processing
	Cores/Threads	4/8 or higher	Concurrent process handling
Random Access Memory (RAM)	Capacity (GB)	8 or higher	Storage for contracts, 3D models, and React data
Graphics Processing Unit (GPU)	Type	Dedicated or Integrated	3D modeling rendering, system performance
	VRAM (GB)	2 or higher	Storage for 3D model data
Storage	Type	SSD or HDD	Storage for OS, applications, contracts, models, and data
	Capacity (GB)	256 or higher	Adequate storage capacity
Internet Connection	Speed (Mbps)	10 or higher	Stable and fast connection for blockchain interaction

Table 4.1: Hardware resources

4.4 SOFTWARE RESOURCES REQUIRED

Software Resource	Version
Solidity	0.8.10 or higher
Truffle / Hardhat	Latest version
Ganache / Local Node	7.0.0 or higher
MetaMask	17.0.2 or higher
React.js	VS Code with
Node.js	VS Code with Solidity extensions
Integrated Development Environment (IDE)	
Ethers.js / Web3.js	Latest stable version
IPPS Client	5.4.0, Chai, Jest
Testing Frameworks	Mocha, Chai, Jest
Security Tools	Slither, MythX
Version Control System	Git
Deployment Tools	Docker (optional)

Table 4.2: Software resources

Chapter 5

SOFTWARE REQUIREMENT SPECIFICATION

5.1 INTRODUCTION

The SmartShelf initiative seeks to revolutionize the current method of digital library management. By providing a decentralized and immutable ledger for book ownership and transaction data, it aims to increase transparency, efficiency, and security in the management of digital and physical library resources, reducing piracy and unauthorized access. This project addresses critical challenges in digital publishing and library systems by combining blockchain's inherent properties with modern technologies, offering a transformative solution for authors, readers, educational institutions, and publishers seeking a secure and reliable platform for book distribution and ownership verification.

5.1.1 Purpose and Scope of Document

To provide readers with a comprehensive understanding of the blockchain-based SmartShelf initiative. It serves as a reference for stakeholders, authors, readers, and developers to align their expectations and understanding by outlining the project's objectives, scope, deliverables, and key considerations. The document covers the project overview, goals, stakeholders, deliverables, implementation strategy, timeline, constraints, assumptions, risks, and communication methods related to the development and deployment of the decentralized digital library system.

5.1.2 Overview of responsibilities of Developer

The developer for the blockchain-based SmartShelf system is responsible for creating and implementing smart contracts using Solidity, integrating them with the React.js front-end application, and ensuring the system operates securely and efficiently. They collaborate with the team to deliver a seamless user experience, perform thorough testing to identify and fix bugs, follow industry best practices, and adhere to coding standards to provide a robust and reliable decentralized digital library solution.

5.2 USAGE SCENARIO

A user who wants to access or purchase a digital book is a usage scenario for the SmartShelf system using blockchain technology. They open the web application built with React.js, set up a secure account using MetaMask for authentication, and browse or select the desired book. The system uses smart contracts to verify the transaction, record ownership, and distribute royalties to the author automatically. This ensures the integrity of digital content transactions and creates an immutable, transparent record of book ownership and access history.

5.2.1 User profiles

User Profile	Description
Author	An individual or entity that creates and wishes to publish digital content. They interact with the system to upload their work, set a price, and view their sales and earnings.
Reader/ Buyer	An individual who wishes to purchase, own, and read digital content. They interact with the system to browse the library, purchase books, and access their collection of owned content.
Platform Administrator	A role responsible for maintaining the front-end application, curating featured content, and managing platform-level parameters (though not controlling user transactions).
System Auditor/ Reviewer	Monitors transactions, verifies smart contract executions, and ensures transparency and accuracy in book ownership records and royalty distributions.

Table 5.1: User Profile

5.2.2 Use Cases

Use Case	Description	Actors	Assumptions
Publish Book	Authors can publish their digital books securely and transparently on the blockchain, creating a unique token for each.	Author	The system assumes the author has the legal rights to the content they are publishing.
Purchase Book	Readers can purchase ownership of a book using cryptocurrency. The transaction is recorded on the blockchain	Reader, Author	The system assumes the reader has a valid digital wallet with sufficient funds. The smart contract handles the fund transfer and ownership change
Access Content	Readers can verify their ownership via their wallet to access and read the content they have purchased.	Reader	The system assumes the data on the blockchain is accurate and can verify the reader's ownership token.

Table 5.2: Use Cases

5.2.3 Use Case View

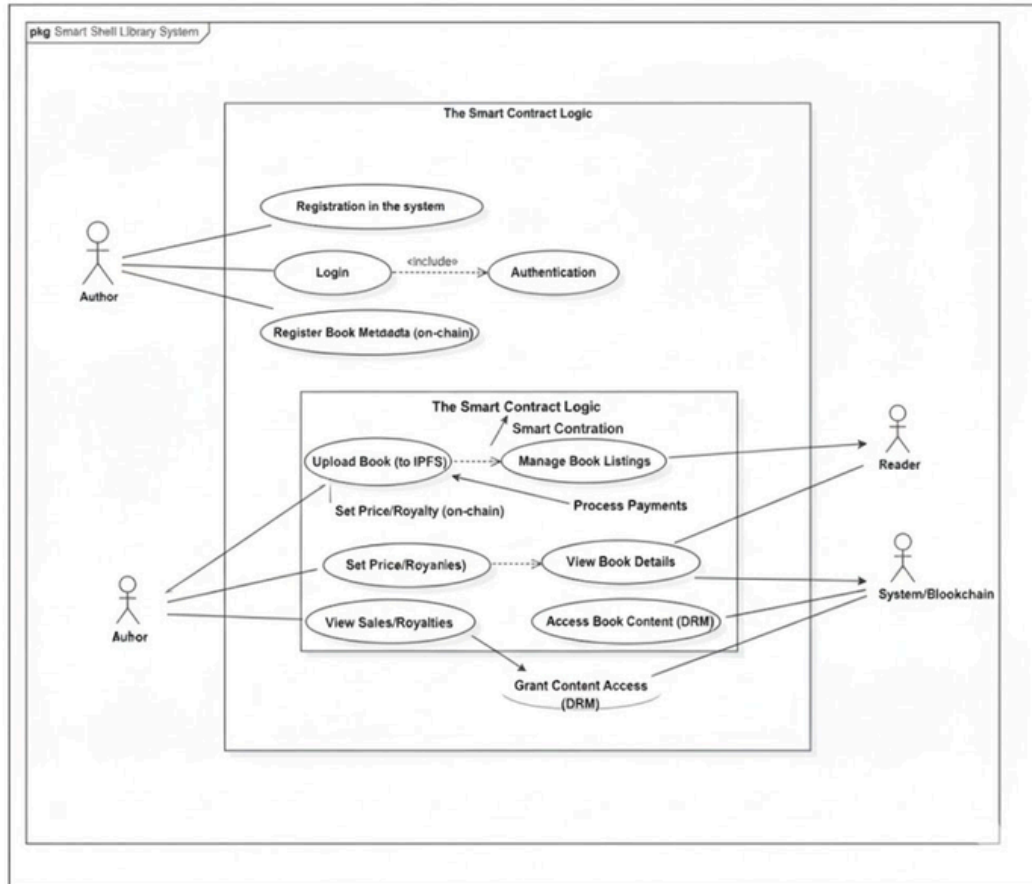


Figure 5.1: Use case diagram

5.3 Data Model and Description

5.3.1 Data Description

The SmartShelf blockchain-based digital library system project's dataset consists of information on digital book details (title, author, description, price, file hash/IPFS link), user accounts and identities, transaction history (purchases, borrowing, returns), and royalty distribution records. In order to guarantee transparency, accuracy, and secure ownership, all data is stored on a decentralized ledger, ensuring immutable records that can be verified by both authors and readers

Data	Description
Book Ownership Details	Information about book ownership, including the owner's wallet address and the unique token ID (NFT).
Transaction Records	Records of book purchases, including buyer and seller addresses, date, and transaction amount (price).
Book Metadata	Data describing the book, including title, author name, description, cover image URL (linked from IPFS).
User Identity Information	Public wallet addresses of authors and readers involved in the transactions, ensuring secure and authenticated interactions.

Table 5.3: Data Description

5.3.2 Data objects and Relationships

The SmartShelf blockchain-based digital library system's data objects include information on digital books, author and reader identities, transaction records, and royalty distribution details. These objects are linked through relationships such as book-to-author, book-to-user (ownership), and transaction-to-book, forming a secure and transparent data ecosystem that ensures accurate tracking of book ownership, access history, and royalty payments.

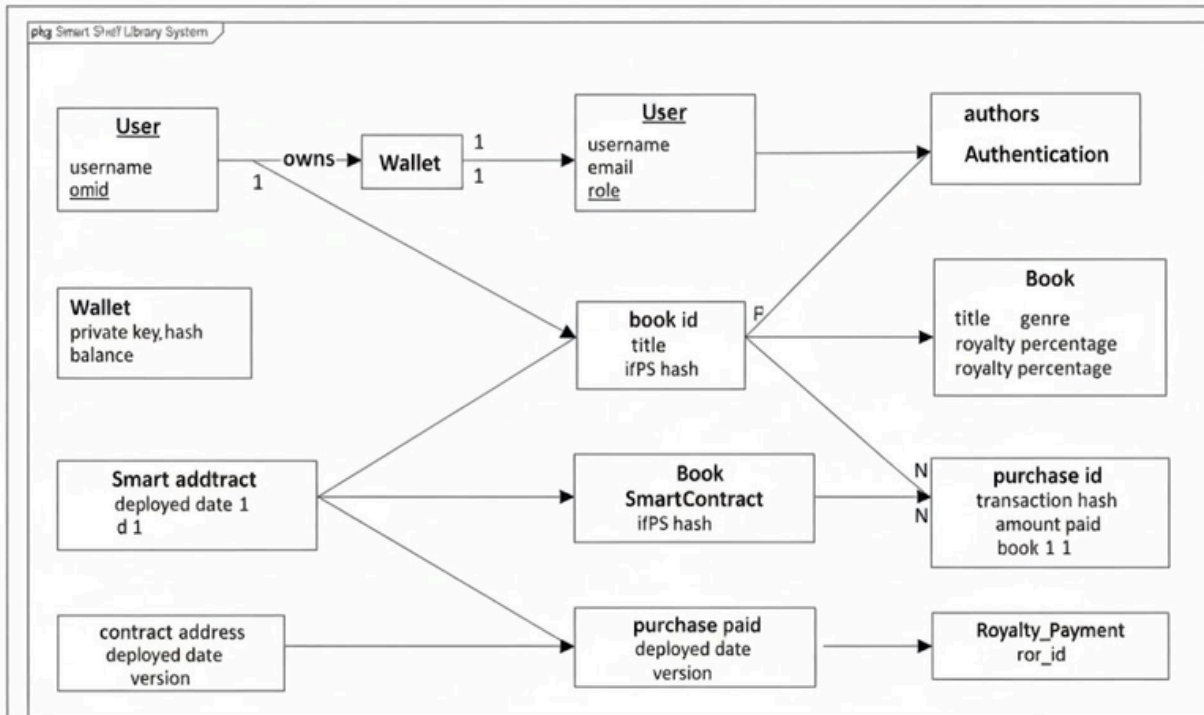


Figure 5.2: Smart Shell Decntelized Library System ER Diagram

5.3.3 Data Flow Diagram

Level 0 Data Flow Diagram

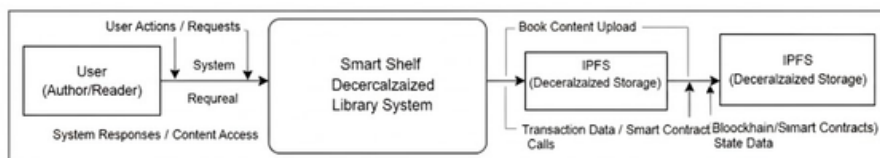


Figure 5.3: Level 0: 0 Flow Diagram

Level 1 Data Flow Diagram

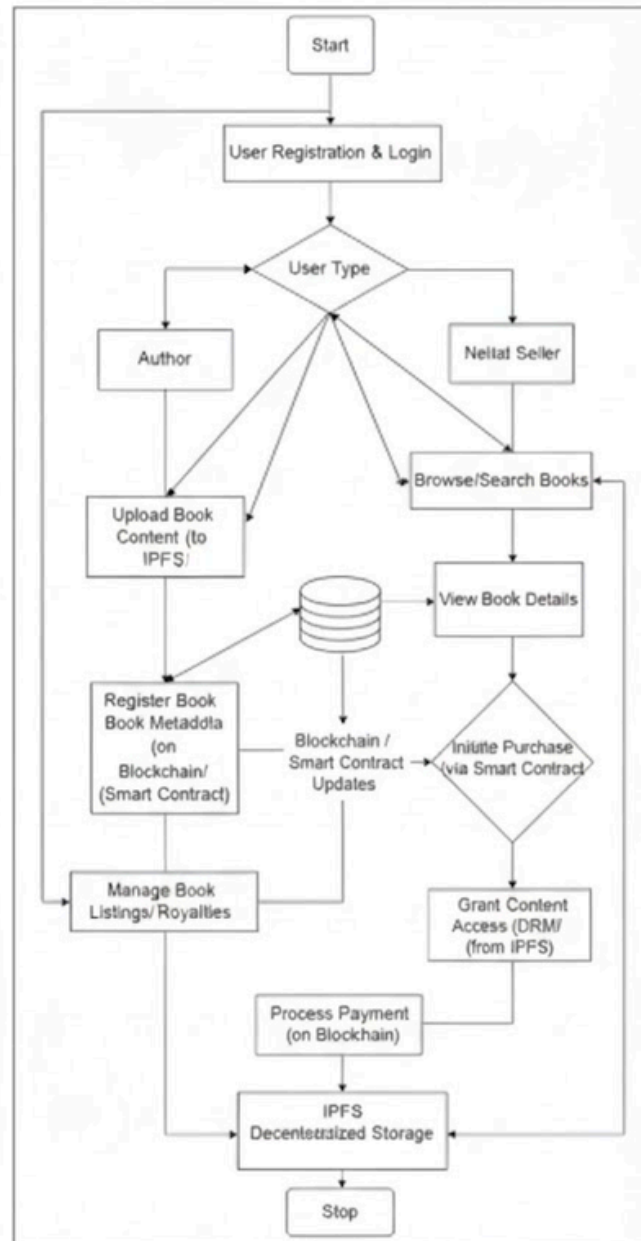


Figure 5.4: Level 1 Data Flow Diagram

5.3.4 Activity Diagram:

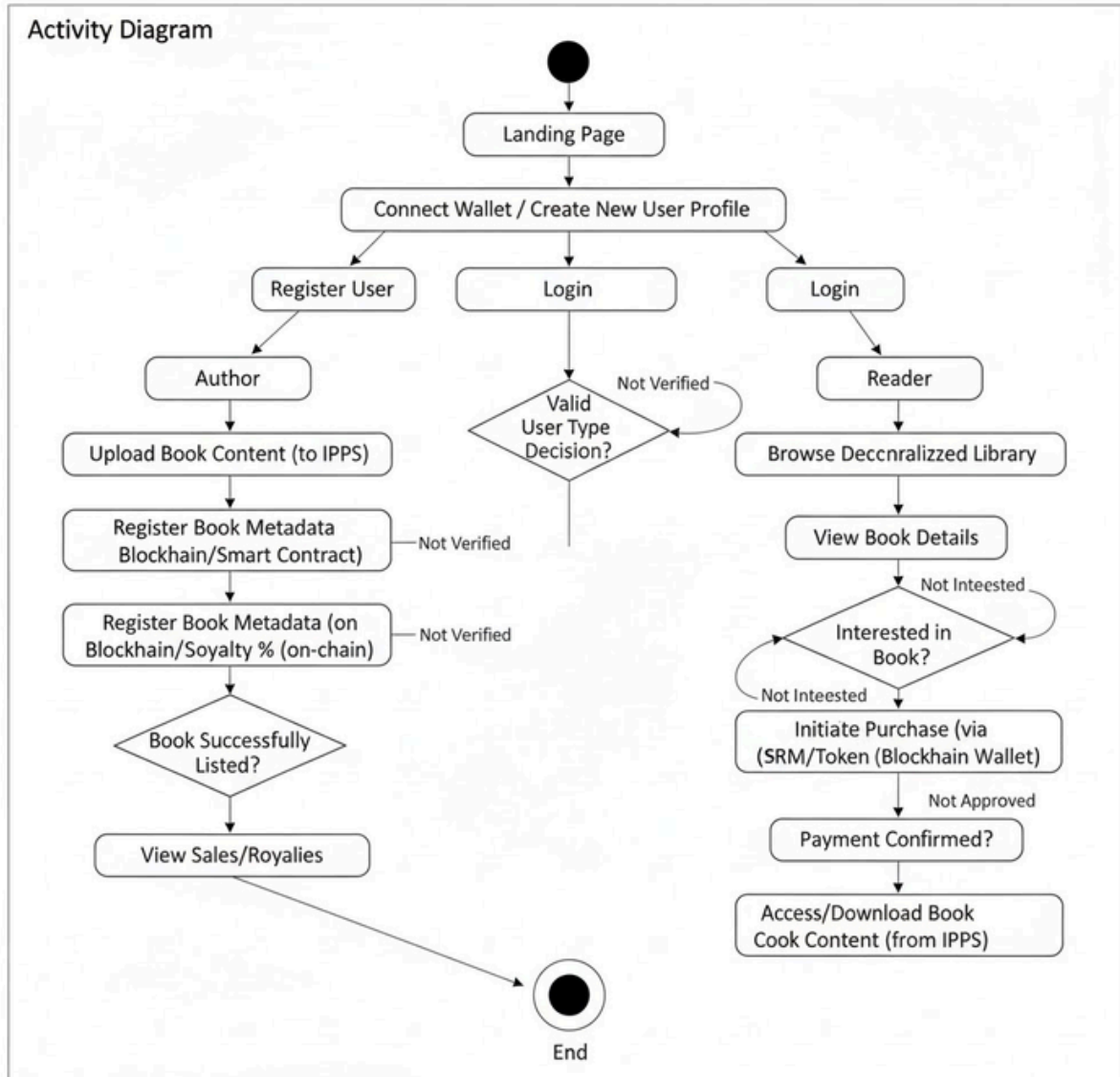


Figure 5.5: Activity Diagram

5.3.5 Non Functional Requirements

- **Security:** To safeguard sensitive data, prevent unauthorized access, and maintain the integrity of the blockchain, the system should ensure strong security measures. This covers safe key management, encryption, and authentication techniques for authors, readers, and transactions.

- **Scalability:** Without sacrificing performance, the system should be able to handle a high volume of book uploads, purchases, borrowings, and royalty transactions. To accommodate growing libraries and an expanding user base, it should scale efficiently.
- **Reliability:** The system must be highly dependable to guarantee accurate and timely recording of book ownership, transactions, and royalty distribution. It should provide reliable data storage and retrieval while minimizing downtime.
 - **Transparency:** The blockchain-based system should make digital book ownership and transaction records transparent, making it easy for authors, readers, and administrators to verify and trace data. It should ensure an auditable record of all activities involving digital content.
- **Usability:** The system should feature an intuitive user interface that allows authors, readers, and administrators to navigate and interact easily. It should provide clear instructions, simple procedures, and helpful feedback to ensure a positive user experience.
- **Interoperability:** The system should work with existing library databases, e-learning platforms, and digital content management tools. It should provide smooth integration with external systems and data sources for seamless operation.
 - **Performance:** In terms of response times, transaction processing, and data retrieval, the system should perform efficiently. Peak loads should be handled effectively, and it should continue operating satisfactorily even under heavy usage.
- **Compliance:** The system should follow all applicable copyright laws, digital content regulations, and blockchain standards. Compliance is essential to maintain the trust and confidence of stakeholders, authors, and readers.
 - **Disaster Recovery:** To minimize the risk of data loss and ensure continuity in the event of system failures or unforeseen circumstances, the system should have reliable backup and recovery procedures. A clear disaster recovery plan should be in place.
 - **Auditability:** To track and monitor activities related to book ownership, transactions, and royalty payments, the system should provide thorough auditing capabilities. It should include comprehensive logging and reporting for verification and accountability.

- **Compliance:** The system should follow all laws and regulations pertaining to blockchain technology, data protection, copyright, and digital content management. To maintain the trust and confidence of authors, readers, and other stakeholders, it should ensure compliance with relevant laws and standards.
- **Disaster Recovery:** To minimize the risk of data loss and guarantee continuity in the event of system failures or unforeseen circumstances, the system should have reliable backup and recovery processes. It should include a clear and actionable disaster recovery strategy for digital books, ownership records, and transaction histories.
- **Auditability:** To track and monitor operations related to book ownership, purchases, borrowing, and royalty payments, the system should provide comprehensive auditing capabilities. It should include detailed logging and reporting functionalities for verification, accountability, and dispute resolution.

5.3.6 State Diagram

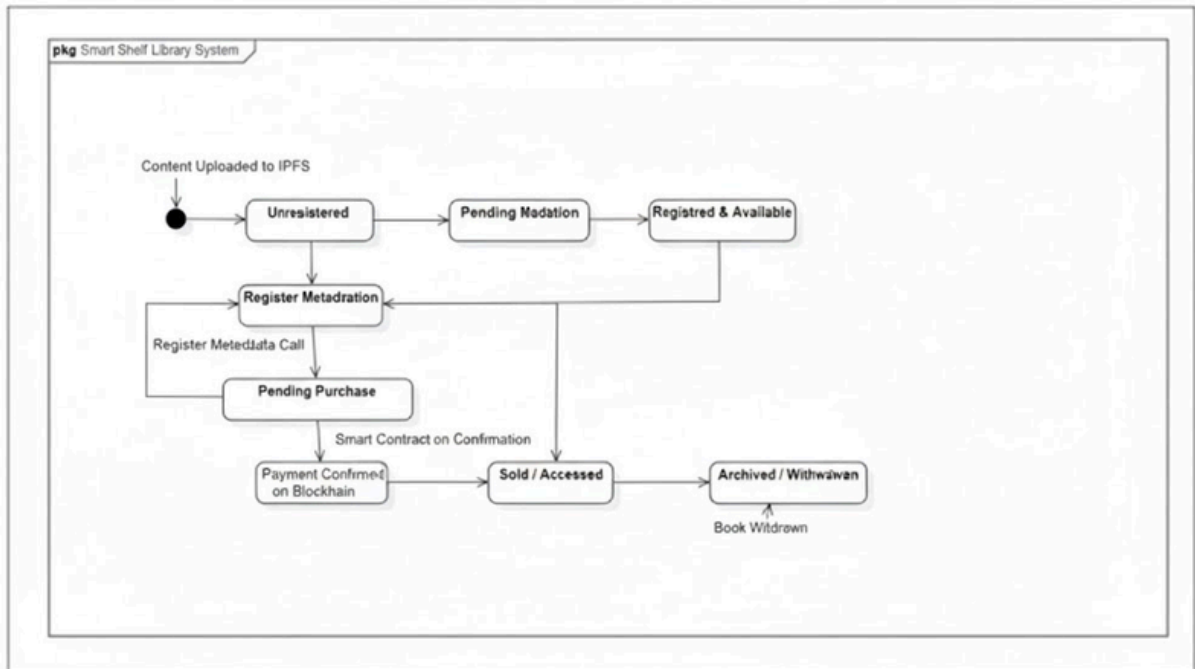


Figure 5.6: State Chart Diagram

5.3.7 Design Constraints

- **Regulatory Compliance:** The SmartShelf system must comply with all applicable laws and regulations, including those related to copyright, digital content distribution, data privacy, and blockchain technology.
- **Interoperability:** To ensure data consistency and compatibility, the system should be developed to seamlessly interact with existing library databases, e-learning platforms, and other relevant digital content management systems.

CHAPTER 6. DETAILED DESIGN DOCUMENT USING APPENDIX A AND B

5.3.8 Software Interface Description

The SmartShelf system interacts with multiple external software systems and interfaces to ensure secure, seamless, and reliable management of digital library resources. These interfaces include:

- **Blockchain Network Interface:** SmartShelf communicates with the Ethereum blockchain (or any chosen blockchain network) to record book ownership, track transactions, and execute smart contracts for royalty distribution. This ensures immutability, transparency, and secure verification of all transactions.
- **Digital Wallet Integration:** The system interfaces with user wallets such as MetaMask or other Web3 wallets for authentication, secure transaction signing, and management of user accounts. This allows authors and readers to interact with the system safely without exposing private keys.
- **Content Storage Interface (IPFS / Decentralized Storage):** SmartShelf connects with decentralized storage solutions like IPFS to store digital book files securely. The system references these files via hashes on the blockchain, ensuring authenticity and preventing unauthorized duplication.
- **Front-End Web Interface (React.js / Web3 UI):** The React.js-based front-end communicates with smart contracts via Web3.js or Ethers.js libraries. It allows authors to upload content, manage sales, and view royalties, while readers can browse, purchase, borrow, and access their owned books.
- **External APIs and Library Databases:** SmartShelf can integrate with existing digital libraries, educational platforms, or publishing APIs to synchronize metadata, user accounts, or verify ISBN/book details. This ensures interoperability and consistent data across platforms.
- **Logging and Auditing Interfaces:** The system interfaces with internal logging and reporting modules that track every transaction, smart contract execution, and user activity for auditing purposes. These logs are accessible to authorized administrators to maintain transparency and accountability.

Overall, these software interfaces work together to provide a secure, decentralized, and user-friendly digital library system that ensures transparent ownership, automated royalty distribution, and seamless access to digital content.

6.1.1 Class Diagram

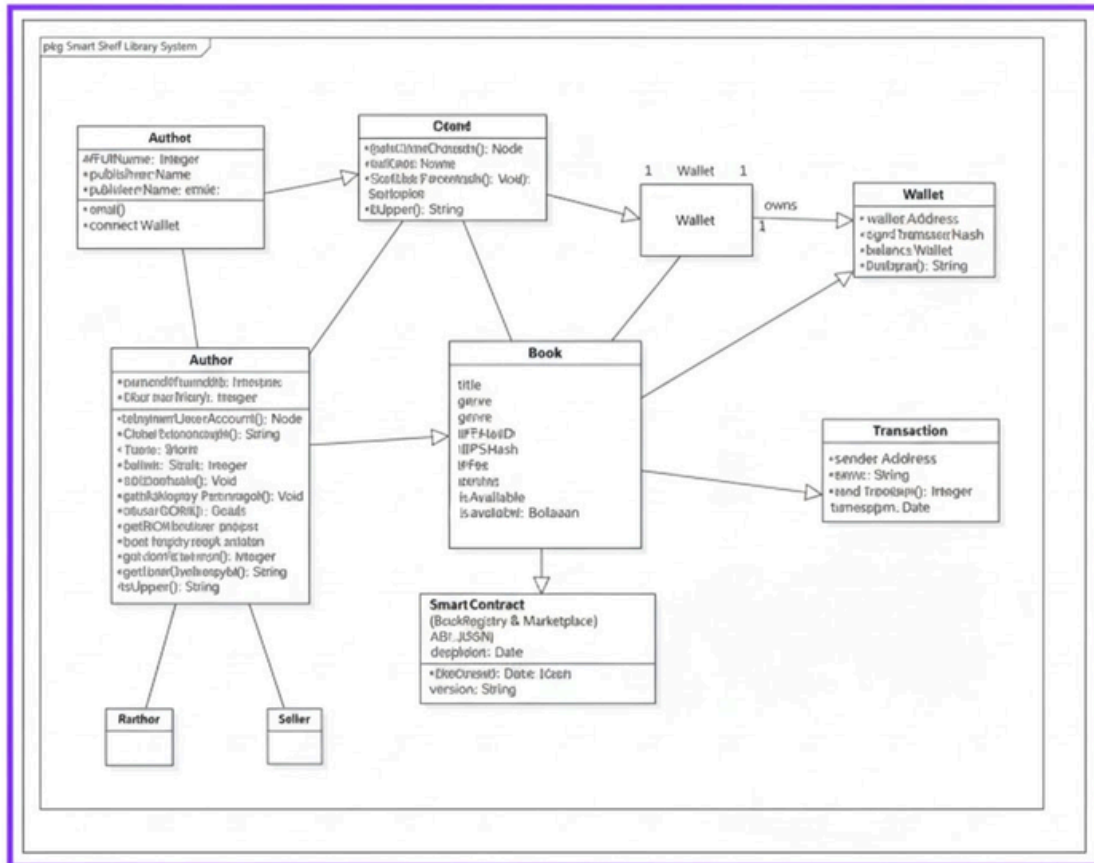


Figure 6.2: Class diagram

CHAPTER 7.

PROJECT IMPLEMENTATION

Project Implementation Plan for SmartShelf

1. **Requirement Analysis:**

Gather and finalize the functional and non-functional requirements for SmartShelf. Identify stakeholders, define user roles (authors, readers, admins), and outline system specifications, including blockchain, storage, and UI/UX requirements.

2. **System Design & Architecture:**

Design the overall architecture of SmartShelf, including blockchain network selection, smart contract structure, decentralized storage (IPFS), front-end design with React.js, and integration points for wallets and APIs.

3. **Smart Contract Development:**

Develop and deploy smart contracts using Solidity to handle book ownership, borrowing, purchases, and royalty distribution. Ensure contracts are secure, efficient, and tested for potential vulnerabilities.

4. **Front-End Development:**

Build a responsive, intuitive web interface using React.js. Implement user authentication, book browsing, purchasing, and access dashboards for authors and readers. Integrate Web3.js/Ethers.js for blockchain interactions.

5. **Decentralized Storage Integration:**

Connect SmartShelf with decentralized storage solutions like IPFS to securely store digital book files. Ensure data integrity by referencing content hashes in smart contracts.

6. **Testing & Quality Assurance:**

Perform unit testing for smart contracts, integration testing for front-end and blockchain interactions, and usability testing for the interface. Identify and fix bugs, optimize performance, and validate security measures.

7. **Deployment & Launch:**

Deploy smart contracts on the selected blockchain testnet and mainnet. Host the front-end application and configure wallet integrations. Launch the system to a limited user base for initial feedback.

8. **Monitoring, Maintenance & Updates:**

Continuously monitor system performance, user activity, and smart contract execution. Address any technical issues, update features based on user feedback, and ensure compliance with security standards.

Chapter 8

SOFTWARE TESTING

8.1 Types of Tests Performed

- Smart Contract Tests
- Blockchain Integration Tests
- User Interface Tests
- Integration Tests
- Security Tests
- Performance Tests
- Transaction Flow Tests
- Usability Tests
- Regression Tests
- Deployment and Environment Tests

8.2 Test Cases & Test Results

Test ID	1 Truffle/Hardhat Installation
Test Description	Successful installation of Truffle framework
Input Data	installation of Truffle framethetwad
Expected Data	Successful installation of Truffle mr framwork, command
Actual Data	command 'truffle version' runs
Status	

Table 8.1: Test Results - Test ID 1

Test ID	2 Ganache Setup
Test Description	Local blockchain network on Ganachain network
Input Data	running on Ganache network
Expected Data	http: funde accounts with http:/127.0.7545 were accounts
Actual Data	
Status	

Table 8.2: Test Results - Test ID 2

Test ID	2 Solidity Contract Deployment Smart contract files
Test Description	Successful deployment of contracts on the blockchain
Input Data	Successful deployment of contracts on the blockchain
Expected Data	Ganache accerhain
Actual Data	
Status	

Table 8.2: Test Results - Test ID 2

Test ID	5 React.js Integration with Web.js fon Web3.js/Ethers.js
Test Description	Successful integration of React.js for user interface
Input Data	Successful deployment of contracts on user interface
Expected Data	Contracts deployed suddrsd of tollus, and read dallly
Actual Data	Successful integgrailly and fetchas from the book list
Status	intetrace

Table 8.3: Test Results - Test ID 2

8.2 Test Cases & Test Results

Test ID	5 Metamsak Integration
Test Description	Metamsak plugin
Input Data	Successful integration of Metamsak
Expected Data	Successful integration of Metamsak for blockchain interaction
Actual Data	
Status Pass	Successful integration or integration

Table 8.5: Test Results - Test ID 5

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